

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CODING GOLF COMPETITION

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO4 – Articulation (BL3)	Assessment:	Assessment Tool 1 – Coding Golf Competition
Weightage:	35% of CIA	Total Marks:	100
CO4 Statement:	Apply hashing strategies for efficient data storage and sorting algorithms for arrangement of data.		
Student Name:	S.Roshni	Reg. No.:	192525185
Section / Batch:	AI&ML	Date:	17/08/2024

Code of Conduct

I S.Roshni(192525185)_____ (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.Roshni_____

Instructions

- Log in to the assigned coding platform - Code chef (<https://www.codechef.com/skill-test/c-language-dsa>). Total: 100 Marks.
- You can only attempt this assessment once. Also, you will not be able to pause the assessment after starting.
- You are not allowed to switch tabs during the assessment, otherwise you will be blocked from the assessment.
- You will get a detailed report on your performance at the end of the assessment.

Section / Topic	Focus	Q Nos	Marks
Data structures	Code Chef	28	100
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Accurately interprets all problem requirements and constraints before coding.	Understands most requirements with minor omissions.	Partially understands the problem; misses some constraints.	Misinterprets the problem or ignores key requirements.
Correctness of Solution	30	Program compiles successfully and passes all hidden and visible test cases.	Program passes most test cases with minor errors.	Program passes some test cases but has logical issues.	Program fails to compile or produces incorrect results.
Data Structure & Algorithm Selection	20	Selects the most appropriate data structure and algorithm with an optimized solution.	Chooses an appropriate solution with minor inefficiencies.	Uses a workable but less efficient approach.	Uses an inappropriate data structure or algorithm.
Code Quality & Documentation	20	Well-structured, modular, readable, and properly indented code with meaningful identifiers.	Readable code with minor formatting issues.	Basic coding style with inconsistent formatting.	Poorly organized code that is difficult to understand.
Time & Space Optimization	10	Demonstrates excellent optimization and correctly analyses time and space complexity.	Good optimization with minor inefficiencies.	Limited optimization and incomplete complexity analysis.	No optimization and incorrect or missing complexity analysis.

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

☰

Saved

🌞 📶 SHAIK ROSHNI

01 Hrs : 23 Min < Prev 1 / 30 Next >

Statement

How do you print the number 108 using arithmetic operations on the numbers 9 and 12 in C?

Did you like the problem?
507 users found this helpful

👍

🗨

💬

Select one of the following options:

☐ print(9/12);

☐ print(9*12);

☐ cout<<9*12;

☒ printf("%d", 9*12);

Submit

Next

☰

✓

🌞 📶 SHAIK ROSHNI

01 Hrs : 23 Min < Prev 2 / 30 Next >

Statement

IDE

C

```
1 #include <stdio.h>
2
3 int main() {
4     printf("$");
5     return 0;
6     // your code goes here
7 }
8
9
10
```

Submit

Next

01 Hrs : 24 Min

< Prev

3 / 30

Next >

Response Saved

Statement

MCQ

What will be the output of the given code if the user enters 'Hello World!'?

☒ Enter a sentence: Hello World! You entered: Hello World!

☐ Hello World! Enter a sentence: You entered: Hello World!

☐ Enter a sentence: You entered: Hello World!

☐ You entered: Hello World!

Submit

Next

Not Attempted

SHAIK ROSHNI

01 Hrs : 20 Min

< Prev 4 / 30 Next >

Statement

Write a C program that takes two floating-point numbers as input and calculates their product. Print the result with two decimal places.

Sample 1:

Input	Output
0.2 2.1	Product: 0.42

More Info

Time limit
1 secs

Memory limit
1.5 GB

Source Limit
50000 Bytes

Did you like the problem?
105 users found this helpful

C

```
1 #include <stdio.h>
2
3 int main() {
4     float a,b;
5     scanf("%f %f",&a,&b);
6     printf("%.2f",a*b);
7     return 0;
8 }
9
10
```

Submit

Next

Write a C program that takes a three-digit integer as input and prints its digits in reverse order (With a string message and spaces between the digits. See the sample output).

Sample 1:

Input	Output
456	Digits in reverse order: 6 5 4

More Info

Time limit

1 secs

Memory limit

1.5 GB

Source Limit

50000 Bytes

Did you like the problem?

80 users found this helpful



```
1 #include <stdio.h>
2
3 int main() {
4     int n,reverse=0,remainder;
5     scanf("%d",&n);
6     while(n!=0)
7     {
8         remainder = n % 10;
9         reverse=reverse*10+remainder;
10        n=n / 10;
11    }
12    printf("%d ",reverse);
13    return 0;
14    // your code goes here
15
16
17 }
18
19
```

Test against Custom Input

456

Run

Submit

Next

Not Attempted

SHAIK ROSHNI

01 Hrs : 10 Min < Prev 6 / 30 Next >

Statement

```
#include <stdio.h>

int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);

    if (num % 2 == 0) {
        printf("%d is even.\n", num);
    } else {
        printf("%d is odd.\n", num);
    }

    return 0;
}
```

What is the purpose of the if/else construct in the given code ?

☐ Check if the number is positive

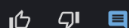
☐ Check if the number is zero

☒ Check if the number is even

☐ Check if the number is negative

Did you like the problem?

94 users found this helpful



Submit

Next

35°C

ENG

13:39

Not Attempted

SHAIK ROSHNI

01 Hrs : 05 Min < Prev 7 / 30 Next >

Statement

Write a C program that takes three integers as input and determines the largest among them. Print the largest number.

Sample 1:

Input	Output
1 2 3	Largest: 3

More Info

Time limit
1 secs

Memory limit
1.5 GB

Source Limit
50000 Bytes

Did you like the problem?
168 users found this helpful

C

```
1 #include <stdio.h>
2
3 int main() {
4     int n,digit,largest = 0;
5     scanf("%d",&n);
6     while (n!=0)
7     {
8         digit = n % 10;
9         if(digit > largest)
10            largest = digit;
11         n=n/10;
12     }
13     printf("%d",largest);
14     return 0;
15 }
16
17
18
```

SubmitNext

Not Attempted

SHAIK ROSHNI

01 Hrs : 04 Min < Prev 8 / 30 Next >

Statement

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char exampleUsername[] = "aValid1";
    int result = (strlen(exampleUsername) >= 8 && isalnum(exampleUsername[0]) &&
        isdigit(exampleUsername[strlen(exampleUsername) - 1]));

    printf("%d\n", result);

    return 0;
}
```

Did you like the problem?
42 users found this helpful

What will be the output of this code?

☐ 0

☐ 1

☐ -1

☐ 2

SubmitNext

Statement

```
#include <stdio.h>

int main() {
    int x = 5;
    int y = 10;
    printf("Sum: %d", x + y);
    return 0;
}
```

What will be the output of the given code ?

☐ Sum: 15

☒ Sum: -5

☐ Compilation Error

☐ No output

Not Attempted

SHAIK ROSHNI

01 Hrs : 03 Min

Response Saved

Statement

```
#include <stdio.h>

main() {
    printf("Hello, World!");
}
```

What will be the output of the given code ?

☒ Hello, World!

☐ Compilation Error

☐ Hello

☐ World

Did you like the problem?

21 users found this helpful

Like

Dislike

Comment

Submit

Next

Not Attempted

SHAIK ROSHNI

01 Hrs : 02 Min

Response Saved

Statement

```
#include <stdio.h>

int main() {
    int numbers[3] = {1, 2, 3};
    printf("%d", numbers[3]);
    return 0;
}
```

What will be the output of the given code ?

☐ 0

☐ 2

☒ Garbage Value

☐ Compilation Error

Did you like the problem?

Like

Dislike

Comment

```
#include <stdio.h>

int main() {
    int values[4] = {10, 20, 30, 40};
    printf("%d", values[4]);
    return 0;
}
```

What will be printed by the given code ?

☐ 40

☐ 0

☒ Garbage Value

☐ Compilation Error

SubmitNext

Statement

Write a C program that takes input for an array of 8 integers and finds the maximum value in the array. Print the maximum value.

Input Format

A single line containing 8 space-separated integers.

Sample 1:

Input	Output
1 3 2 4 6 5 7 8	8

More Info

Time limit
1 secs

Memory limit
1.5 GB

Source Limit
50000 Bytes

Did you like the problem?
35 users found this helpful

C

```
5 for(i =0;i < 8;i++){
6     scanf("%d",&a[i]);
7     max = a[0];
8     for(i=1;i<8;i++){
9         {
10             if(a[i]> max)
11                 max = a[i];
12         }
13     }
14     printf("%d",max);
15     return 0;
16 }
17
```

Test against Custom Input

1 3 2 4 6 5 7 8

Sample Input

1 3 2 4 6 5 7 8

Your Output

8

RunSubmitNext

Statement

```
#include <stdio.h>

int main() {
    int num = 1, sum = 0;
    while (num <= 5) {
        sum += num;
        num++;
    }
    printf("%d", sum);
    return 0;
}
```

What will be printed by the given code ?

☐ 10

☒ 15

☐ 5

☐ 25

SubmitNext

Response Saved

Statement

```
#include <stdio.h>

int subtract(int a, int b) {
    printf("Result: %d\n", a - b);
}

int main() {
    subtract(10, 5);
    return 0;
}
```

What will be printed by the main function?

☒ Result: 5

☐ Result: 10

☐ 5

☐ Compilation Error

Statement

```
#include <stdio.h>

int main() {
    int values[] = {2, 3, 4, 5, 6};
    int key = 4;

    for (int i = 0; i < 5; i++) {
        if (values[i] > key) {
            break;
        }
        printf("%d ", values[i]);
    }

    return 0;
}
```

What will be printed by the loop?

☐ 2 3

☒ 2 3 4

☐ 2 3 4 5

☐ 2 3 4 5 6

Did you like the problem?

14 users found this helpful



Submit

Next

←

https://www.codechef.com/skill-test/LCAS3/problems/LCAS138

🔍 ⭐ ⚙️ 👤 ... Chat

☰ ● Not Attempted 🌞 📶 SHAIK ROSHNI 53 Min : 31 Sec

✔️ Response Saved

Statement

```
#include <stdio.h>

int sumArray(int arr[], int size) {
    int sum = 0;
    for (int i = 0; i < size; i++) {
        sum += arr[i];
    }
    return sum;
}

int main() {
    int numbers[] = {1, 2, 3, 4, 5};
    int result = sumArray(numbers, 5);
    printf("%d\n", result);
    return 0;
}
```

What will be printed by the main function?

☐ 10

☒ 15

☐ 20

☐ 25

Did you like the problem?
13 users found this helpful

👍 🗨️ 💬

Submit

Next

☰ ● Not Attempted 🌞 📶 SHAIK ROSHNI 52 Min : 11 Sec

✔️ Response Saved

Statement

```
#include <stdio.h>

int main() {
    int t;
    scanf("%d", &t);

    while (t--) {
        int num;
        scanf("%d", &num);
        printf("Square: %d\n", num * num);
    }

    return 0;
}
```

How many numbers should be given by the user to make this code run?

☐ t-1

☐ t

☒ t+1

☐ Any

Did you like the problem?
23 users found this helpful

👍 🗨️ 💬

Submit

Next

35°C 13:57

SHAIAK ROSHNI

50 | Min : 36 | Sec < Prev 21 / 30 Next >

Statement

Which of the following is NOT a valid data type in C?

Select the correct option.

☐ float

☐ char

☒ string

☐ double

Did you like the problem?

104 users found this helpful

Submit

Next

SHAIAK ROSHNI

48 | Min : 27 | Sec < Prev 22 / 30 Next >

Statement

Given the height (1.82 meter) and weight (72 kg) of the Chef, calculate his BMI (Body Mass Index).

Formula to calculate the BMI:

BMI = $\frac{\text{Weight (in kilograms)}}{\text{Height}^2 \text{ (in meters)}}$

Sample 1:

Input

Output

21.736506

More Info

Time limit

1 secs

Memory limit

1.5 GB

Source Limit

50000 Bytes

Did you like the problem?

141 users found this helpful

C

```
1 #include <stdio.h>
2
3 int main() {
4     float height = 1.82;
5     float weight = 72;
6     float bmi = weight / (height * height);
7     printf("%f\n",bmi);
8     return 0;
9 }
10
```

Sample Input

Your Output

21.736506

Submit

Next

Not Attempted

SHAIK ROSHNI

45 | Min : 02 | Sec < Prev 24 / 30 Next >

Statement

Write a C program to modify the given string "RoadChef" to "CodeChef" and print the updated string.

Sample 1:

Input	Output
	CodeChef

More Info

Time limit
1 secs

Memory limit
1.5 GB

Source Limit
50000 Bytes

Did you like the problem?
35 users found this helpful

C

```
1 // Update the string and print
2 #include <stdio.h>
3
4 int main() {
5     char str[] = "RoadChef";
6     str[0]='C';
7     str[1]='o';
8     str[2]='d';
9     str[3]='e';
10    printf("%s\n",str);
11
12
13    return 0;
14 }
15
```

Submit

Next

Not Attempted

SHAIK ROSHNI

47 | Min : 02 | Sec < Prev 25 / 30 Next >

Statement

What does the function `strstr()` do in C?

Did you like the problem?
39 users found this helpful

Select one of the following options:

☐ Compares two strings

☐ Finds the first occurrence of a substring in a string

☐ Splits a string into substrings

☐ Counts the occurrences of a character in a string

Submit

Next

Response Saved

Not Attempted

SHAIK ROSHNI

39 Min : 49 Sec Prev 26 / 30 Next

Statement

Write a program using `while` loop to check if the positive number N is prime or not.

Note: Prime numbers are the numbers that are divisible by only 1 and the number itself.

Input Format

- The first and only line of input will contain a single positive number N .

Output Format

- Output on a single line, `YES` if N is prime, else `NO`.

Constraints

- $0 \leq N \leq 100000$

Sample 1:

Input	Output
4	NO

Explanation:

4 is divisible by 2 also apart from 1 and 4

Sample 2:

Input	Output
-------	--------

C

```
1 #include <stdio.h>
2
3 int main() {
4     int num;
5
6     scanf("%d", &num);
7     if(num <= 1){
8         printf("NO\n");
9         return 0;
10    }
11    int i = 2;
12    int isprime = 1;
13    while(i < num){
14        if(num%i == 0){
15            isprime = 0;
16            break;
17        }
18        i++;
19    }
20    if (isprime){
21        printf("YES\n");
22    }
23    else{
24        printf("NO\n");
25    }
26
27    return 0;
28
29 }
```

Test against Custom Input

4

Run Submit Next

Not Attempted

SHAIK ROSHNI

33 Min : 03 Sec Prev 27 / 30 Next

Statement

Write a program to check whether a number can be express as sum of two [prime numbers](#).

Input Format

- The first line contains an integer, T , denoting the number of test cases.
- Each test contain a single positive integer N - the number that needs to be checked.

Output Format

- For each test case, print:
 - `YES` if N can be expressed as sum of two primes.
 - `NO` if not.

Constraints

- $1 \leq T \leq 10$
- $1 \leq N \leq 1000$

Sample 1:

Input	Output
3	YES
7	YES
19	NO
11	

Explanation:

C

```
1 #include <stdio.h>
2
3
4 int isprime(int n) {
5     if(n<2) return 0;
6     for(int i=2;i<=n;i++)
7         if(n%i==0) return 0;
8     return 1;
9 }
10 int main(){
11     int t,n;
12     scanf("%d",&t);
13     while(t--){
14         scanf("%d",&n);
15         int ok = 0;
16         for(int i = 2; i<=n/2;i++){
17             if (isprime(i) && isprime(n-i)){
18                 ok = 1;
19                 break;
20             }
21         }
22         printf(ok ? "YES\n": "NO\n");
23     }
24     return 0;
25 }
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Not Attempted

SHAIK ROSHNI

27 Min : 09 Sec

Prev 28 / 30 Next

Statement

Write a program to check if the given number N is Armstrong number.

Note: Armstrong number is a number that is equal to the sum of its own digits each raised to the power of the number of digits.

Input Format

- The first and only line of input contains a single integer, N - The number that needs to be checked.

Output Format

- Print on a single line:
 - YES if N is Armstrong number.
 - NO otherwise.

Constraints

- $0 \leq N \leq 1000000$

Sample 1:

Input	Output
153	YES

Explanation:

$1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$.

The number 153 has 3 digits, so each digit was raised to the power of 3 before summing.

C

```
1 #include <stdio.h>
2 #include <math.h>
3 void isArmstrongNumber(int num) {
4     int temp = num, digits = 0, sum = 0;
5     while (temp > 0){
6         digits++;
7         temp /= 10;
8     }
9     if(num==0)digits = 1;
10    temp = num;
11    while (temp > 0){
12        int rem = temp % 10;
13        sum+= pow(rem,digits);
14        temp /=10;
15    }
16    if(sum == num)
17        printf("YES\n");
18    else
19        printf("NO\n");
20 }
21
22
23
24
25 int main() {
26     int n;
27     scanf("%d", &n);
28
29     isArmstrongNumber(n);
30 }
```

Test against Custom Input

153

Run

Submit

Next

Statement

Write a C program to find the maximum and minimum elements in an array using pointers.

Input Format

- The first line of input contains a single integer N , denoting the number of elements in the array.
- Second line of input contains N space separated integers, the elements of the array.

Output Format

- Output on a single line, the max value in the array.

Sample 1:

Input	Output
5 3 4 1 2 3	4

More Info

Time limit
1 secs

Memory limit
1.5 GB

Source Limit
50000 Bytes

C

```
3 // Function to find maximum element in an array using pointers
4 void findMax(int arr[], int size, int *max) {
5     *max = *arr;
6     for(int i = 1; i < size; i++){
7         if(*(arr+i) > *max){
8             *max = *(arr + i);
9         }
10    }
11 }
12 // Complete the function
13
14 }
15
16 int main() {
17     int n;
18     scanf("%d", &n);
19     int array[n];
20     for(int i = 0; i < n; i++){
21         scanf("%d", &array[i]);
22     }
23     int max;
24
25     // Call the findMaxMin function
26     findMax(array, n, &max);
27
28     printf("%d\n", max); // Output the max element
29 }
```

Test against Custom Input

5
3 4 1 2 3

Sample Input

Run

Submit

Next

Not Attempted

SHAIK ROSHNI

22 Min : 22 Sec

Prev 30 / 30 View Summary

Statement

How do you access members of a structure in C?

☐ Using dot operator (.)

☐ Using arrow operator (->)

☒ Using asterisk operator (*)

☐ Using ampersand operator (&)

Did you like the problem?

18 users found this helpful

Like

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Report

Submit

View Summary

C language online test – Report

Next attempt again after 29 Days

Share test score

View Solutions

Your Total Score

2300 /3000 points

23 Correct Answers

05 Wrong Answers

02 Unattempted

76%

SKILL SCORE

Topic wise summary

Topic	Correct	Total	Remarks	Resources
Output & Printing in C	5	5	Good	Learn →
User Inputs in C	1	3	Weak	Learn →
Conditionals in C	2	3	Average	Learn →
Debug common errors in C	1	3	Weak	Learn →
Arrays in C	2	3	Average	Learn →
Loops using C	3	3	Good	Learn →